Inference and effect size

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.73743    0.08499   20.44   <2e-16 ***
WBC         -0.36085    0.01243  -29.03   <2e-16 ***
```

> Calculate a 99% confidence interval for the change in the log odds of dengue associated with a one-unit increase in WBC.

$$\hat{\beta}_1 \pm z^* SE_{\hat{\beta}_1}$$

$$-0.361 \pm 2.58\,(0.012)$$

$$= (-0.392, -0.330)$$

Activity

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.73743    0.08499   20.44   <2e-16 ***
WBC         -0.36085    0.01243  -29.03   <2e-16 ***
```

> How could we calculate a 99% confidence interval for the change in the *odds* of dengue associated with a one-unit increase in WBC?

$$\left(e^{-0.392}, e^{-0.330}\right) = (0.676, 0.719)$$

Activity

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.73743    0.08499   20.44   <2e-16 ***
WBC         -0.36085    0.01243  -29.03   <2e-16 ***
```

99% confidence interval in terms of odds: (0.675, 0.720)

> How do I interpret this confidence interval?

We are 99% confident that a one-unit increase in WBC is associated with a change in the odds of dengue by a factor of between 0.675 and 0.720

99% confident: imagine doing this experiment many times, with many different datasets. For each dataset, I calculate a 99% confidence interval. 99% of those intervals contain the true β_1

Activity

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Coefficients:
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(Intercept)  1.73743    0.08499   20.44   <2e-16 ***
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```

99% confidence interval in terms of odds: (0.675, 0.720)

> Your friend, seeing your confidence interval from question 2, states that there is a 1% chance that the true change in odds associated with a one-unit increase in WBC is less than 0.675 or greater than 0.720. Is your friend correct?

No — e^{β_1} is either in interval (0.675, 0.720), or it isn't. (we don't know) But e^{β_1} is not random — there is no probability involved with this specific interval

Types of inference questions

- Is there a relationship between these variables?
  - Hypothesis testing
- What is the size of relationship between these variables?
  - Confidence intervals

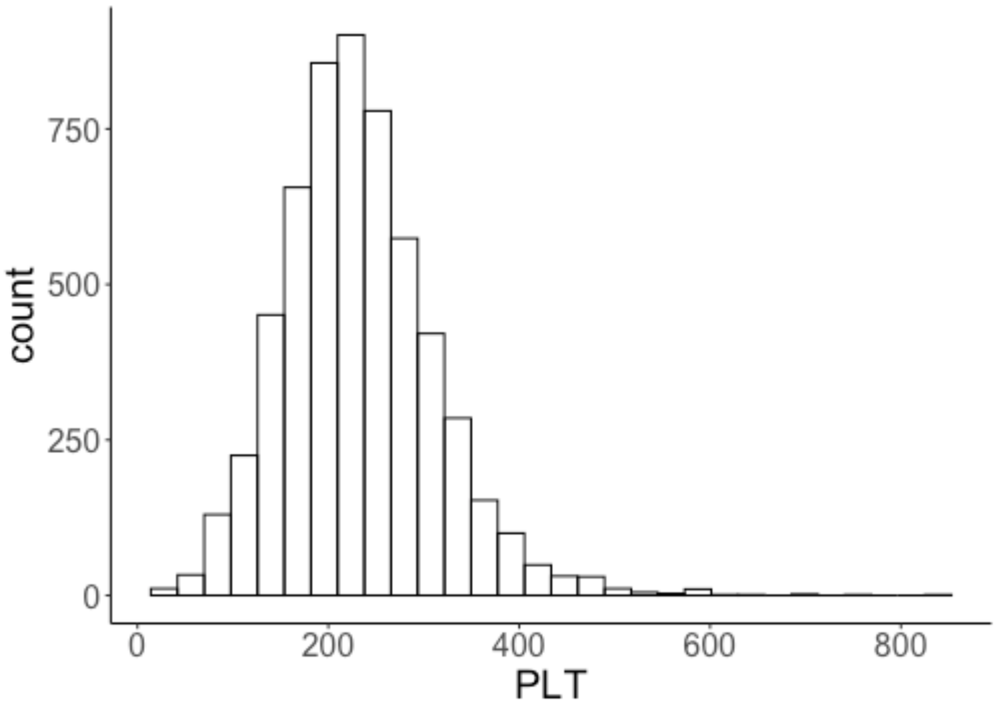
Today: thinking about effect size

Thinking about effect size

```
              Estimate Std. Error z value Pr(>|z|)    
(Intercept)  1.2525593  0.1548038   8.091  5.9e-16 ***
Age          0.1383186  0.0099763  13.865  < 2e-16 ***
WBC         -0.2523294  0.0135371 -18.640  < 2e-16 ***
PLT         -0.0060276  0.0006113  -9.860  < 2e-16 ***
```

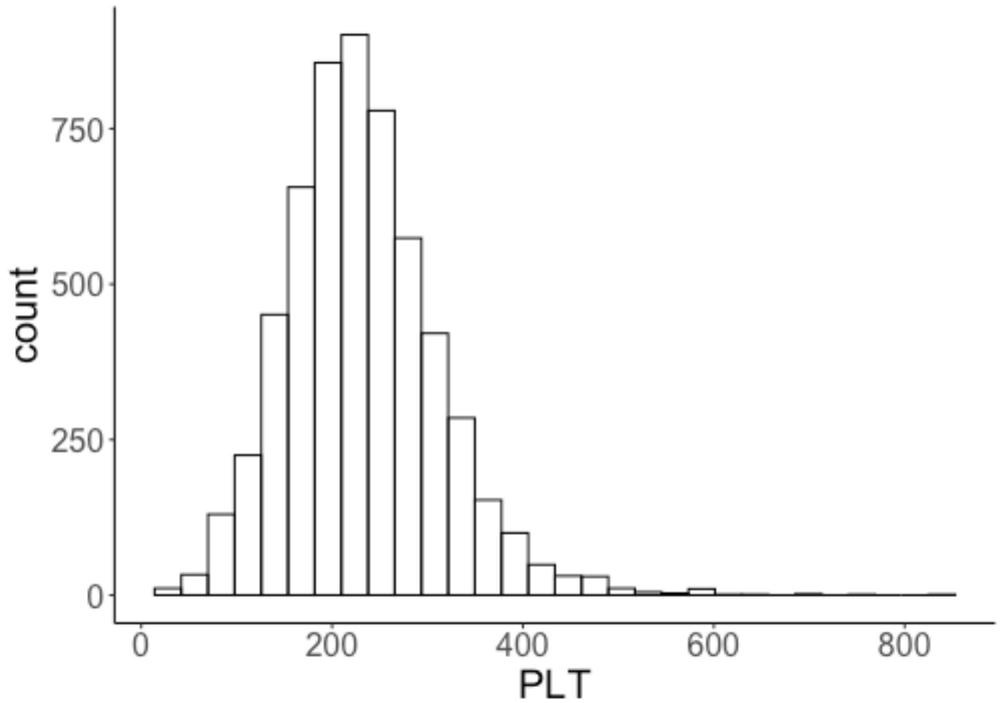
> Your friend claims that the coefficient on platelet count (PLT) is very close to 0, so platelet count is not a meaningful variable to include in the model. Is your friend correct?

Thinking about effect size



- The coefficient on PLT (-0.006) is interpret in terms of a one-unit increase
- The scale of PLT ranges from 0 to 800 -- a one-unit increase is very small!

Thinking about effect size



> If a one-*unit* increase has different meaning across different variables, how else could we think about the effect size of a variable?

Effects in terms of standard deviations

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.2525593  0.1548038   8.091  5.9e-16 ***
Age          0.1383186  0.0099763  13.865  < 2e-16 ***
WBC         -0.2523294  0.0135371 -18.640  < 2e-16 ***
PLT         -0.0060276  0.0006113  -9.860  < 2e-16 ***
```

The *standard deviation* of PLT in the data is 80.02.

> What is the change in log-odds associated with a one *standard deviation* in PLT, holding the other variables constant?

-0.006 : change in log-odds for one-unit increase in PLT

$-0.006(80.02) = -0.482$ change for a one-SD increase in PLT

Effects in terms of standard deviations

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.2525593  0.1548038   8.091  5.9e-16 ***
Age          0.1383186  0.0099763  13.865  < 2e-16 ***
WBC         -0.2523294  0.0135371 -18.640  < 2e-16 ***
PLT         -0.0060276  0.0006113  -9.860  < 2e-16 ***
```

The *standard deviation* of PLT in the data is 80.02.

Holding Age and WBC constant, a one *standard deviation* increase in PLT is associated with:

- A decrease of 0.482 in the log-odds of dengue
- A change in the *odds* by a factor of $e^{-0.482} = 0.617$ (i.e., a decrease of almost 40%)!

Effects in terms of standard deviations

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.2525593  0.1548038   8.091  5.9e-16 ***
Age          0.1383186  0.0099763  13.865  < 2e-16 ***
WBC         -0.2523294  0.0135371 -18.640  < 2e-16 ***
PLT         -0.0060276  0.0006113  -9.860  < 2e-16 ***
```

The *standard deviation* of PLT in the data is 80.02.

The estimated probability of dengue at the mean values of Age, WBC, and PLT is about 0.2. If we fix Age and WBC:

- Probability when PLT = mean + 1 sd is 0.135
- Probability when PLT = mean - 1 sd is 0.292

Activity

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Activity

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.507329   0.311939  -8.038 9.14e-16 ***
loss        -0.846499   0.238335  -3.552 0.000383 ***
smoke100    -0.137151   0.035816  -3.829 0.000128 ***
hlthplan     0.613613   0.051233  11.977  < 2e-16 ***
age         -0.012148   0.001043 -11.652  < 2e-16 ***
height       0.056845   0.004520  12.577  < 2e-16 ***
```

The standard deviation of the `loss` variable is 0.074.

> How do the odds of exercise change with a one standard deviation increase in loss, holding the other variables constant?

Activity

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.507329   0.311939  -8.038 9.14e-16 ***
loss        -0.846499   0.238335  -3.552 0.000383 ***
...           ...         ...        ...      ...
```

The standard deviation of the `loss` variable is 0.074.

> Want to test whether a one-standard deviation increase in `loss` is associated with a decrease in the odds of at least 5% (holding other variables constant).